

ASSIGNMENT #2 (DMWH) Marks: 64

Section A ($5 \times 2 = 10$ Marks)

Q1. [CO3] What is the need for Data Preprocessing?

Answer:

Data preprocessing is essential because raw data is often **incomplete, noisy, inconsistent, or redundant**. Preprocessing improves the quality of data and ensures accurate analysis.

Reasons for Data Preprocessing:

1. **Handle Missing Data:** Replace or estimate missing values.
2. **Reduce Noise:** Smooth out errors or outliers.
3. **Data Integration:** Combine data from multiple sources.
4. **Data Transformation:** Normalize and scale data for algorithms.
5. **Reduce Redundancy:** Remove duplicate or irrelevant data.

Example: In a customer database, missing ages or inconsistent address formats are corrected during preprocessing.

Q2. [CO3] Explain different types of Data Repositories.

Answer:

Data repositories are storage systems where data is maintained. They include:

1. **Operational Databases:** Used for daily operations (OLTP).
 - *Example:* Banking transactions database.
 2. **Data Warehouses:** Consolidated storage for historical and analytical purposes (OLAP).
 - *Example:* Retail warehouse storing 5 years of sales data.
 3. **Data Marts:** Smaller subsets of data warehouses, focused on specific departments.
 - *Example:* Sales data mart for marketing team.
 4. **Transactional Databases:** Store live transaction records.
 - *Example:* Online shopping order database.
 5. **Metadata Repositories:** Stores information about data (data about data).
 - *Example:* Schema of tables, data types, constraints.
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Q3. [CO3] State the different layers of Data Warehouse.

Answer:

A data warehouse has the following layers:

1. **Data Source Layer:** Original sources like ERP, CRM, and external data.
 2. **Data Staging Layer:** ETL process for cleaning, transforming, and preparing data.
 3. **Data Storage Layer:** Fact and dimension tables stored in warehouse or OLAP cubes.
 4. **Data Presentation Layer:** Provides tools for querying, dashboards, and reporting.
 5. **Metadata Layer:** Information about warehouse structure, data definitions, and lineage.
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Q4. [CO3] What is the difference between Data Warehousing and Data Mining?

Feature	Data Warehousing	Data Mining
Purpose	Store and consolidate data	Discover hidden patterns and insights
Data	Historical and integrated	Used for prediction and knowledge discovery
Process	ETL (Extract, Transform, Load)	Algorithms like classification, clustering
Output	Organized data	Patterns, rules, predictions
Example	A warehouse storing 10 years of sales	Predicting which customers may churn

Q5. [CO3] What are the issues in data integration?

Answer:

Data integration combines data from multiple sources, but faces several issues:

1. **Schema Integration:** Different databases may have incompatible schemas.
 2. **Data Conflicts:** Conflicting data values between sources.
 3. **Redundancy:** Duplicate data increases storage and affects analysis.
 4. **Data Quality:** Inconsistent, missing, or incorrect data.
 5. **Semantic Heterogeneity:** Same data may have different meanings in different systems.
 6. **Scalability:** Integrating large datasets efficiently.
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Section B (3 × 7 = 21 Marks)

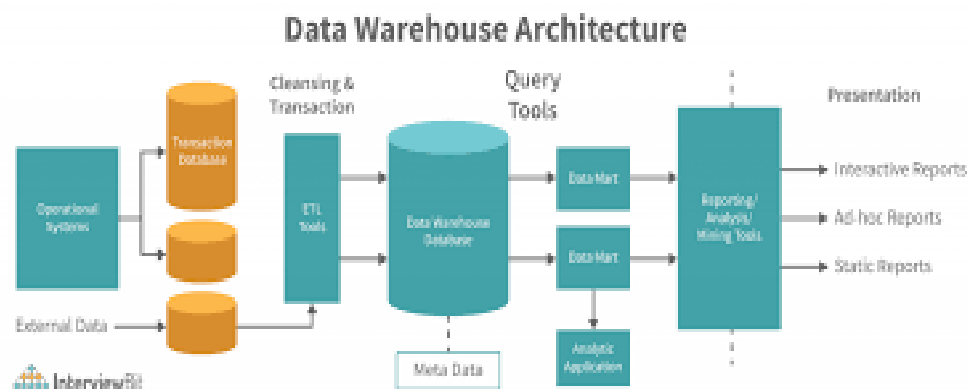
Q1. [CO3] Discuss the architecture of Data Warehouse.

Answer:

Data Warehouse Architecture consists of several layers:

1. **Data Sources:**
 - Internal (ERP, CRM) and external sources.
2. **ETL (Extract, Transform, Load):**
 - Extract data → Clean → Transform → Load into warehouse.
3. **Data Storage:**
 - Fact tables (quantitative data) and dimension tables (contextual data).
4. **Metadata Layer:**
 - Describes data structure, source, and rules.
5. **Presentation/Access Layer:**
 - Provides tools for analysis, OLAP, dashboards, and reports.

Diagram:



Q2. [CO3] Explain the architecture of Data Mining with diagram.

Answer:

Data Mining System Architecture has these components:

1. **Database/Data Warehouse:** Source of integrated data.
2. **Data Mining Engine:** Performs algorithms for classification, clustering, regression, and association.
3. **Pattern Evaluation Module:** Evaluates discovered patterns for usefulness.
4. **User Interface:** Interactive tools for querying and visualization.
5. **Knowledge Base:** Domain knowledge guiding the mining process.

Diagram:



Q3. [CO3] What are the strategies of data reduction?

Answer:

Data reduction reduces volume without losing significant information. Strategies include:

1. **Dimensionality Reduction:** Reduce number of attributes.
 - Example: PCA (Principal Component Analysis).
2. **Numerosity Reduction:** Use models or aggregation to represent data.
 - Example: Regression, histograms.
3. **Data Compression:** Encode data efficiently.
 - Example: Run-length encoding, wavelet transforms.
4. **Data Cube Aggregation:** Aggregate data for OLAP queries.
5. **Sampling:** Select representative subset of data for analysis.

Example: Using 10% of customer transactions instead of entire dataset for trend analysis.

Section C (3 × 11 = 33 Marks)

Q1. [CO3] Define KDD. Identify and describe phases of KDD.

Answer:

KDD (Knowledge Discovery in Databases) is the process of finding **valid, novel, and useful patterns** from large datasets.

Phases of KDD:

1. **Data Selection:** Choose relevant data from various sources.
2. **Data Preprocessing:** Clean data, remove noise, handle missing values.
3. **Data Transformation:** Normalize, aggregate, or format data for mining.
4. **Data Mining:** Apply algorithms to extract patterns (classification, clustering).
5. **Pattern Evaluation:** Identify interesting and valid patterns.

6. **Knowledge Presentation:** Visualize and present insights to users.

Example: E-commerce uses KDD to find which products are often bought together.

Q2. [CO3] Illustrate Data Preprocessing in detail.

Answer:

Data preprocessing ensures data quality for accurate mining.

Steps of Data Preprocessing:

1. **Data Cleaning:**
 - Handle missing values, smooth noisy data, remove duplicates.
 - *Example:* Replace missing age with mean age.
 2. **Data Integration:**
 - Combine data from multiple sources.
 - *Example:* Merge CRM and sales database.
 3. **Data Transformation:**
 - Normalize, aggregate, or discretize data.
 - *Example:* Scale salary from 0–1 for analysis.
 4. **Data Reduction:**
 - Reduce volume using sampling, aggregation, or dimensionality reduction.
 5. **Data Discretization:**
 - Convert continuous attributes into discrete intervals.
 - *Example:* Age → Young (0–25), Adult (26–50), Senior (>50).
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Q3. [CO3] Why do we process data? Highlight the importance of each stage in production.

Answer:

Data processing ensures high-quality, consistent, and usable data for analysis.

Importance of Each Stage:

1. **Data Collection:** Gather relevant raw data from sources.
 - Importance: Provides the foundation for analysis.
2. **Data Cleaning:** Remove errors and inconsistencies.
 - Importance: Prevents inaccurate results.
3. **Data Integration:** Combine multiple datasets.
 - Importance: Provides a holistic view of information.

4. **Data Transformation:** Convert data to suitable formats.
 - Importance: Makes data compatible with mining algorithms.
5. **Data Reduction:** Reduce volume while retaining meaningful information.
 - Importance: Improves efficiency and reduces computational cost.
6. **Data Mining:** Extract meaningful patterns.
 - Importance: Converts data into actionable knowledge.
7. **Knowledge Presentation:** Visualize and communicate findings.
 - Importance: Supports decision-making and strategic planning.

Example: In a manufacturing company, processed data helps predict machine failures, reduce downtime, and optimize production.
